Classical Field Theory — Exercise Sheet 13

Radiation estimates and boundary conditions

Hints and answers

Use Lecture 13's real Maxwell fields, retarded power theorem and amplitude $\mathcal{A}(u,n) = (4\pi)^{-1}\int \mathbf{J}(u+n\cdot y, y)\mathrm{d}^3y$. Here $n\in\mathbb{S}^2$, $u=t-r$, and $P_{\mathrm{T}}(n)=I-nn^{\mathsf{T}}$. The third problem also uses Lecture 6's time-reflected advanced kernel. The singular-source formula in the last problem is supplied separately.

Exercise 13.1 (A general slow-time source). Let $\rho_0(s,y)$ be scalar and $J_0(s,y)$ vector-valued, both real and smooth with compact spacetime support, satisfying $\partial_s\rho_0 + \operatorname{div}_y J_0 = 0$. For $0<\varepsilon\le 1$ put

$$\rho_\varepsilon(t,y)=\rho_0(\varepsilon t,y),\qquad \mathbf{J}_\varepsilon(t,y)=\varepsilon J_0(\varepsilon t,y),\qquad p_0(s)=\int y\rho_0(s,y)\mathrm{d}^3y.$$

Define finite constants

$$M_0=\int \sup_s|\partial_s J_0(s,y)|\mathrm{d}^3y,$$
$$M_1=\int |y|\sup_s|\partial_s^2 J_0(s,y)|\mathrm{d}^3y,$$
$$M_2=\frac12\int |y|^2\sup_s|\partial_s^3 J_0(s,y)|\mathrm{d}^3y.$$

(a) Verify conservation and prove $p_0'(s)=\int J_0(s,y)\mathrm{d}^3y$. Find the scaled dipole $p_\varepsilon(t)=\int y\rho_\varepsilon(t,y)\mathrm{d}^3y$ and its second derivative.

(b) Prove, uniformly for all $u\in\mathbb{R}$, the explicit absolute estimate

$$\left|P_\varepsilon(u)-\frac{|\ddot p_\varepsilon(u)|^2}{6\pi}\right|\le\frac{2M_0M_1+M_1^2}{4\pi}\varepsilon^5.$$

(c) Improve the integrated error to $O(\varepsilon^6)$ by retaining the first delayed-sampling correction. Give an explicit bound using M_0,M_1,M_2. Why does this improvement not assert that the directional power density has an $O(\varepsilon^6)$ error?

Hint. Taylor-expand $\partial_s J_0(\varepsilon u+\varepsilon n\cdot y,y)$ at εu before squaring. The first correction is odd in n, whereas the transverse projector is even.

Answer. The dipole is $p_\varepsilon(t)=p_0(\varepsilon t)$. An $O(\varepsilon^6)$ bound is $\varepsilon^6(2M_0M_2+M_1^2+2M_1M_2+M_2^2)/(4\pi)$. The $O(\varepsilon^5)$ cross term cancels only after angular integration.

Exercise 13.2 (A nonradial source at zero acceleration). Choose a nonnegative $f\in C_c^\infty(B_1(0))$ with $\int f=1$ and nonzero first moment $\mu=\int yf(y)\mathrm{d}^3y$. Let $a\in\mathbb{R}^3\setminus\{0\}$ and choose a smooth compactly supported p_0 equal to as^3 on $|s|\le 1$. Use the conserved dipole source

$$p(t)=p_0(\varepsilon t),\qquad \rho=-p(t)\cdot\nabla f,\qquad \mathbf{J}=\dot p(t)f,\qquad 0<\varepsilon\le 1.$$

Compute the exact retarded power $P(0)$. Show that it is positive although $\ddot p(0)=0$, and evaluate it for $a=e_3$, $\mu=e_3/4$. Explain what this rules out about a relative error estimate. You may use the angular moments

$$\int n_in_j\mathrm{d}\Omega=\frac{4\pi}{3}\delta_{ij},\qquad \int n_in_jn_kn_l\mathrm{d}\Omega=\frac{4\pi}{15}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}).$$

Hint. Every sampled slow time $\varepsilon n \cdot y$ lies in the polynomial region. Keep the first moment of f rather than replacing it by a radial average.

Answer.

$$P(0) = \frac{6\varepsilon^6}{5\pi}\left(2|a|^2|\mu|^2 - (a\cdot\mu)^2\right) > 0.$$

For the specified aligned vectors it is $3\varepsilon^6/(40\pi)$.

Exercise 13.3 (The half-retarded, half-advanced field). For a real smooth compact spacetime conserved source, let $(\phi_{\mathrm{em}}^{\mathrm{ret}}, \mathbf{A}_{\mathrm{ret}})$ be its retarded potentials. Define the advanced potentials by replacing $t - |x-y|$ by $t + |x-y|$ in the same $1/(4\pi|x-y|)$ integrals. Time reflection of Lecture 6's kernel gives these smooth sourced solutions. Let the symmetric potentials be half the sum of the two, and form their Maxwell fields.

(a) At future null infinity, $t = r + u$, show that the advanced field is exactly zero for all sufficiently large r (at each fixed u). Find the symmetric field's limiting outward power in terms of the retarded power.

(b) At past null infinity, $t = v - r$, define

$$\overline{\mathcal{A}}(v,n) = \frac{1}{4\pi}\int \mathbf{J}(v - n\cdot y, y)\mathrm{d}^3y, \qquad \overline{\mathcal{V}}(v,n) = \frac{1}{4\pi}\int \rho(v - n\cdot y, y)\mathrm{d}^3y.$$

Derive the leading advanced electric and magnetic coefficients and the signed outward power of the symmetric field. Primes now mean ∂_v. Does averaging the potentials cancel radiation at future null infinity?

Hint. For the advanced field the source time is $t + |rn - y|$. At fixed t, its leading radial derivative has the opposite sign from the retarded case. Use conservation to relate the two advanced amplitudes.

Answer. Future outward power is $P_{\mathrm{ret}}(u)/4$. The advanced coefficients at past infinity are $E_* = -P_{\mathrm{T}}\overline{\mathcal{A}}'$ and $B_* = +n\times\overline{\mathcal{A}}' = -n\times E_*$. The symmetric past outward power is $-\frac{1}{4}\int |n\times\overline{\mathcal{A}}'|^2\mathrm{d}\Omega$.

Exercise 13.4 (Liénard–Wiechert: a supplied singular-source formula). Let $z\colon \mathbb{R}\to\mathbb{R}^3$ be a smooth complete worldline with $|\dot z(s)| \le v_* < 1$. Fix an event (t,x) with $x \neq z(t)$. Let $Q\in\mathbb{R}$ be its charge. For this problem only, take as given the point-source potential prescription

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi}\int \frac{\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s, \qquad \mathbf{A}(t,x) = \frac{Q}{4\pi}\int \frac{\dot z(s)\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s.$$

The delta integrals are understood locally at their roots. You may use the one-variable identity $\delta(g(s)) = \sum_j \delta(s-s_j)/|g'(s_j)|$ for simple roots.

(a) Prove existence and uniqueness of the retarded root $t = s_{\mathrm{ret}} + |x - z(s_{\mathrm{ret}})|$. Evaluate the two potentials in terms of $R = |x - z(s_{\mathrm{ret}})|$, $n = (x - z(s_{\mathrm{ret}}))/R$ and $v = \dot z(s_{\mathrm{ret}})$.

(b) For $z(s) = (3s/5, 0, 0)$ at $(t,x) = (1,0)$, compute the retarded time, Jacobian and scalar potential. Explain exactly which smooth-source hypotheses fail and why this calculation establishes no self-force law.

Hint. The distance function is v_*-Lipschitz in s. This proves global monotonicity even if $x = z(s)$ at a different time. Only the actual retarded root must have nonzero distance.

Answer.

$$\phi_{\mathrm{em}} = \frac{Q}{4\pi R(1 - n\cdot v)}, \qquad \mathbf{A} = \phi_{\mathrm{em}} v.$$

The example gives $s_{\mathrm{ret}} = 5/8$, $1 - n\cdot v = 8/5$, and $\phi_{\mathrm{em}} = 5Q/(12\pi)$.